Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of an AI based All-in-One Solid-Phase Photonic PCR Platform for Companion Animal Disease Diagnostics			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices and Diagnostic Devices	Molecular Genetic Diagnostic Devices	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The core concept is a **fully integrated, automated solid-phase photonic PCR diagnostic platform** designed specifically for animal health, streamlining the entire workflow **inside a single-use disposable cartridge**. By combining sample preparation, nucleic acid extraction, amplification, and detection into one sealed cartridge, the system eliminates the need for multiple instruments, specialized lab infrastructure, and manual handling steps. This enables high-precision molecular diagnostics at the point of care in veterinary clinics, shelters, farms, mobile practices, and field settings **with faster decision-making, improved outcomes, and better infection control.** **A. Companion Animal & Pet Care Diagnostics** • Rapid in-clinic testing for common and high-impact pet pathogens, supporting same-visit diagnosis and treatment plans. • Improved antimicrobial stewardship to help differentiate likely infectious etiologies and			

	reduces empiric broad-spectrum antibiotic use by enabling pathogen confirmation. • Pre-surgical / immunocompromised patient screening can benefit from the work where it leads to risk reduction for high-consequence infections before procedures or during chemotherapy/steroid therapy. **B. Veterinary Public Health & Livestock/Agriculture** • On-site disease detection in herd/flock settings to prevent outbreaks, red • Use economic loss, and support biosecurity. • Biosecurity and surveillance capacity to supports routine monitoring, especially in resource-limited or geographically remote environments. • Implementation and scale-up feasibility for determining the training needs, per-test operational costs, data reporting workflows, and pathways for integration with veterinary public health agencies. **C. AI-Driven Data Mining** • Capture solid-phase photonic PCR fluorescence signals using a high-resolution camera to build datasets • Validate positive/negative classification accuracy and continuously improve performance via mining/ML models • NPU-based image-readout module - Develop an NPU-powered main hard-board and cartridge/processing unit (CUP) for on-device, real-time high-resolution image interpretation. • Cost-effective result reporting - Replace expensive optical analyzers with a camera-based analysis unit, improving affordability and scalability for field deployment. • Reduce economic losses - Earlier detection and faster response reduce treatment costs, productivity loss, and culling impacts
Rationale for Joint R&D Project	Solid-phase photonic PCR is widely regarded as the standard for highly sensitive and specific pathogen detection, but conventional workflows still rely on multiple manual steps such as sample pretreatment, nucleic acid extraction/purification, reagent setup, amplification, and result interpretation often performed on separate instruments in controlled laboratories, limiting timely point-of-care and field use. To close this gap, an integrated AI data-mining layer is needed to automate interpretation and decision support at the edge: NPU-enabled on-device inference can deliver real-time, low-latency analysis without cloud dependence, while XAI can provide transparent, auditable outputs that build trust; combined with multi-modal data fusion (assay signals + run metadata), robustness across operator/device/environment variability, and drift monitoring, this capability would enable reliable deployment outside the lab. However, such tightly integrated edge NPU + XAI functionality is largely absent in current platforms, making it **a critical and necessary innovation for field-ready diagnostics.** To address these constraints, **sample-to-answer "all-in-one" PCR systems** have emerged. These platforms integrate **lysis/pretreatment, nucleic acid extraction, amplification, and detection** into a **single-use sealed cartridge** operated by a compact reader. Users simply load a raw specimen, and the cartridge's embedded microfluidics and pre-stored reagents automatically execute the workflow and deliver an objective result at the point of care. **However, current all-in-one PCR platforms still face key limitations especially in veterinary clinics and field settings:** 1. **High system complexity and cost** due to reliance on active pumps/valves and pressure-control hardware. 2. **Sensitivity loss and higher invalid rates** from matrix-dependent PCR inhibition and variable extraction efficiency. 3. **Limited specimen/workflow flexibility**, often requiring specific collection media or narrowly

	defined protocols. **Challenges remain in achieving reliable pretreatment–PCR interfacing within an all-in-one architecture**, where incomplete fluidic/thermal coupling degrades end-to-end robustness and delays commercialization. 4. **Operational failure modes** that reduce reliability and user confidence. 5. **Challenges in multiplexing and miniaturization**, while maintaining thermal uniformity, optical stability, and high signal-to-noise. 6. **Challenges in commercialization and economic feasibility** for point-of-care molecular diagnostic platforms, as the **absence of robust AI analytics** and the high cost of optical detection modules hinder affordability and scalable deployment. The proposed technology is inherently interdisciplinary and cannot be effectively realized through a single institutional capability. The key innovation lies in individual modules such as pretreatment, solid-phase photonic PCR chemistry, and pretreatment-interfacing microfluidics solid-phase photonic PCR (interfacing microfluidics). However, major technical gaps remain **in the co-optimization and system-level integration that enables consistent sample-to-answer performance across diverse specimen types and use environments**. The Korea research team has successfully developed and validated solid-phase photonic PCR chips at the component level, demonstrating strong analytical capability. **Yet, translating these chips into routine point-of-care and field use still requires substantial workflow integration.** Without a dedicated interfacing layer, operation often remains partially manual and sensitive to user variability, bubbles, leakage, and cross-contamination risk, limiting practical deployment in real-world veterinary settings. **The proposed Korean-German collaboration project is designed around the principle of complementary excellence, where each partner contributes a distinct core competency that is essential to de-risk integration and accelerate technology readiness.** In this context, **Dr. Natalia Beshchasna's** group **in Fraunhofer IKTS** (Germany) has **deep expertise in advanced microfluidic design and integration**, with demonstrated strengths in reliable microfluidic interfacing, manufacturable bonding/assembly strategies, and robustness engineering for lab-to-field translation. • **Korea team contribution: Pretreatment kit and solid-phase photonic PCR with readers** - sample pretreatment workflows - multi-detection using solid-phase photonic PCR, including design and validation strategies to maintain sensitivity and specificity in multiplex conditions - analytical approaches for interpreting multi-target signals and converting raw fluorescence data into actionable diagnostic outputs • **German team contribution: Automated Microfluidics and Cartridge System Engineering** - automated microfluidic engineering - integration engineering for compatibility with compact reader modules (thermal cycling interfaces, optical detection alignment, and biosafe sealing) - design-for-manufacturing considerations to support scalability, reproducibility, and regulatory-aligned quality control
Key Words	1. sample-to-answer 2. solid-phase photonic PCR 3. interfacing microfluidic system 4. pretreatment 5. AI data mining

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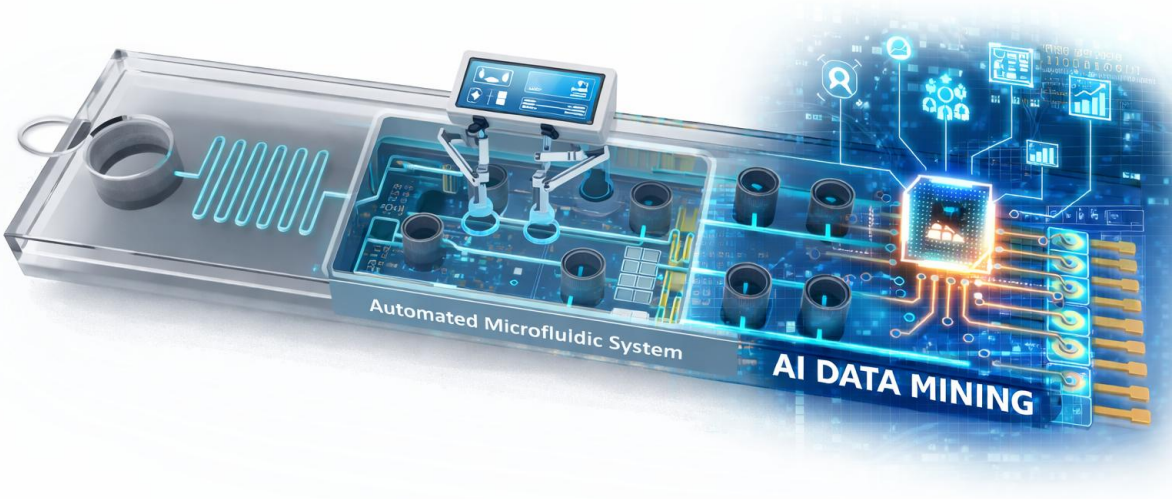
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Development of an integrated diagnostic platform where a user loads a raw or clinical sample into a single-use cartridge and the system automatically performs with the following process: **pretreatment → nucleic acid handling → solid-phase photonic PCR → detection → data mining**  • Integration and interfaces for the sample-to answer one-step cartridge			
R&D Target Specifications	**A. Feline Herpesvirus-1 (FHV-1)** • Clinical relevance: Common cause of feline upper respiratory disease; important for shelter outbreaks and multi-cat environments. • Diagnostic goal: Same-visit confirmation to guide isolation, antiviral consideration, and supportive care. **B. Feline Calicivirus (FCV)** • Clinical relevance: Major contributor to feline respiratory disease; high transmission risk in group housing. • Diagnostic goal: Rapid identification to strengthen infection-control measures and reduce spread. **C. Feline Infectious Peritonitis (FIP)** • Clinical relevance: FIP is a severe condition associated with FCoV infection; early molecular support can aid triage and workup. • Diagnostic goal: - Primary molecular target: Detection of FCoV RNA - Optional advanced target: Detection of FIP-associated viral mutations as adjunctive evidence. • Important limitation to state in proposal: solid-phase photonic PCR findings support diagnosis but may not be definitive for FIP alone; interpret with clinical signs and additional tests.			
R&D Plan	**WP1. System Requirements & Integration Management (Korea + Germany)** • Define system-level specs and manage interfaces to avoid late integration failures. • Define Target Product Profile - sample types, time-to-result, hands-on time, biosafety, output format. • Create Interface Control Documents between extraction chip ↔ interfacing microfluidics ↔			

solid-phase photonic PCR chip/reader

- Define verification plan
  - leak, valve timing, thermal/optical calibration, internal controls, invalid-run criteria.

WP2. Passive-valve Pre-treatment / Nucleic Acid Extraction Chip (Korea)

- Develop a disposable nucleic acid extraction chip that performs lysis → binding → wash → elution using passive valves.
- Optimize extraction chemistry for veterinary matrices
- Build in internal process control
- extraction + inhibition monitoring

WP3. Interfacing Microfluidics (Germany)

- Develop the interfacing microfluidics that physically and functionally connects modules inside one cartridge while maintaining reliability.
- Design the fluidic backbone
- connectors/manifolds, routing channels, seals, vents, bubble management, dead-volume minimization
- Create robust module interfacing
- alignment, docking, pressure/flow window, prevention of backflow/cross-contamination
- Validate manufacturability
- bonding method, assembly flow, leak integrity under thermal cycling

WP4. Solid-phase PCR chips & Reader (Korea)

- Develop solid-phase photonic PCR chips and a dedicated reader to enable rapid, stable, and reproducible PCR detection within a single disposable cartridge.
- Design and fabricate solid-phase photonic PCR chips
- Develop the reader hardware (thermal + optical + mechanical interface)
- Develop control firmware and analysis software
- Verify performance and reliability

WP5. System Integration & Cartridge-Level Verification (Korea + Germany)

- Combination of WP1+WP2+WP3 into a single cartridge workflow and verify end-to-end performance.
- Mechanical/fluidic integration
- End-to-end run scripts and failure detection
- Verification testing
  - hands-on time, sample-to-answer time, leak integrity, carryover/contamination checks.

WP6. Analytical Validation & Veterinary Field Evaluation (Korea + Germany)

- Demonstrate performance for feline targets in relevant settings against reference PCR.
- Analytical
  - LOD, inclusivity/exclusivity, interference, reproducibility
- AI data mining
  - Utilize large-scale fluorescence imaging data from solid-phase photonic PCR to train ML models for improved signal interpretation and higher diagnostic accuracy (SNR enhancement, outlier detection, adaptive thresholds).
- Clinical/field
  - compare to lab PCR using matched specimens
  - measure agreement and time-to-result benefits.
- Usability
  - non-lab operator workflow assessment
  - invalid-rate tracking and root-cause improvements.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Contributions of Fraunhofer** ○ Design and fabrication of the microfluidic cartridge - CAD based design of the microfluidic chip focused on functionality - Chip fabrication via lithography process: cleanroom protocols implemented to fabricate chromium glass mask containing developed design using laser lithography, fabrication of photoresist-based master structure via high resolution UV lithography mask aligner, fabrication of the PDMS flow-cell via soft lithography protocols. - Testing of the microfluidic cartridge in laboratory conditions ○ Development of electrochemical paper-based biosensor for detection of photonic PCR products - Fabrication of paper-based biosensor chip using inkjet printing technology - Development of biosensor assay to provide selective detection of the RNA/DNA of interest and reproducible electrochemical signal. - Developed for the multiplex detection - Selection of surface functionalization method to apply aptamers for the photonic-PCR chip - Integration of the biosensor into photonic PCR chip - Detection of PCR products and biosensor performance
Expected R&R of Korean SME Partner	**Passive-valve Pre-treatment Kits** • Define kit requirements and fluidic workflow • Design and prototype passive valves and microfluidic layout to optimize flow sequencing and leak-proof sealing. • Develop and optimize sample pre-treatment chemistry (lysis, inhibitor handling, extraction yield) for target specimen types. • Verify performance and robustness • Prepare manufacturing transfer package **Solid-phase Photonic PCR Chips & Readers** • Design and fabricate solid-phase photonic PCR chips • Develop the reader: thermal cycling control, optics/sensing, cartridge interface, and calibration. • Build firmware/software for run control and result calling • Integrate chip + reader + assay; tune system-level performance (Limit of detection, repeatability, cycle time) • Execute verification/reliability testing and deliver design documentation for scale-up and validation **AI-enabled data mining & performance optimization** • Build a big-data repository from high-volume, time-series fluorescence image datasets generated by solid-phase Photonic PCR. • Apply machine-learning–based image/signal analytics for sensitive signal extraction and robust artifact/noise rejection • Optimize calling algorithms via adaptive decision-thresholding to improve diagnostic accuracy and robustness across diverse settings.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**I. Technological achievements** Fully integrated sample-to-answer cartridge: • End-to-end workflow in one disposable: sample introduction → lysis → nucleic-acid extraction/purification → solid-phase photonic PCR setup → amplification → detection → result output • Closed system biosafety: leak-proof fluidic architecture and sealed waste chamber to reduce contamination risk and user exposure Microfluidics + reagent stabilization: • Precision microfluidic control: valves/films, capillary structures, or burstable pouches enabling consistent metering and mixing across lots • Pre-stored reagents: dried/lyophilized master mix, primers/probes, controls, and buffers stored on-cartridge with stable shelf life at ambient or refrigerated conditions • On-cartridge quality controls: internal amplification control, process control, and optional calibration features for lot release and field reliability Solid-phase photonic PCR performance targets: • Analytical sensitivity / LOD competitive with lab PCR for relevant targets • Turnaround time optimized via fast thermal transfer and compact reaction volumes • Multiplex capability (e.g., 2–6 targets depending on optics/chemistry) for syndromic panels or variant differentiation • Reproducibility and robustness across temperature/humidity ranges and user types AI data mining performance: • On-device acceleration (NPU): Real-time inference on embedded NPU/edge AI with low latency and reduced cloud dependency • Explainability (XAI): Built-in interpretable outputs to support clinical/operational trust and auditability • Data fusion & context awareness: Multi-modal ingestion (assay signals + metadata such as sample QC, temperature logs, run conditions) to improve accuracy and reduce false calls **II. Commercialization potential** Target market and use cases: • Point-of-care / near-patient diagnostics: clinics, emergency rooms, outpatient centers, smaller hospitals • Public health / border screening: outbreak response kits • Decentralized labs: satellite labs needing rapid PCR without full molecular infrastructure • Non-medical adjacencies: veterinary diagnostics, food safety, environmental monitoring, bio-process QC Differentiators translate to sales: • Ease of use → lower training burden, fewer errors • Lower total cost of ownership: reduced consumables, no separate extraction kits, less rework • Faster time-to-result → clinical decision impact (isolation, antivirals/antibiotics stewardship, throughput) Regulatory and market access pathway (joint Korea–Germany advantage): • Dual-track plan can accelerate global reach: Korea: MFDS pathway and domestic reimbursement/market entry strategy Germany/EU: alignment with EU IVDR expectations (performance evaluation planning, risk management, QMS readiness) Partnership and scale-up readiness: • Strategic suppliers: polymers/films, lyophilization services, optics/filters, enzymes/oligos,

	automated assembly equipment • Pilot production line → validated line → capacity expansion with cost-down roadmap **III. Broader industrial and economic impact** Korea-Germany industrial cooperation: • Combines complementary strengths: Germany: precision engineering, metrology, industrial automation, and deep experience in regulated medical device ecosystems. Korea: electronics integration, high-throughput manufacturing, rapid iteration, and strong biotech/diagnostics commercialization. • Establishes a repeatable collaboration template for future platforms Competitiveness of SMEs and high-value manufacturing: • Creates opportunities for microfluidic tooling/molds, film/adhesive suppliers, automation integrators, reagent formulation and lyophilization specialists. • Moves value upstream into high-margin consumables and platform IP, not just assembly Healthcare system and productivity gains: • Faster diagnosis can reduce unnecessary antibiotics, shorten hospital stays, and limit outbreak spread—benefits that translate into measurable economic savings

6. References

Can be provided upon request.